MILITARY SPECIFICATION

CABLE ASSEMBLY, TEST
(FOR TORPEDO TEST SET MARK 281 MOD 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers a 65-conductor test cable assembly for use with the Torpedo Test Set Mark 281 Mod 1. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-S-6855	Synthetic Rubber Sheets, Strips, Molded or Extruded Shapes
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-W-16878/5	Wire, Electrical, Type EE, 200° C., and 260° C., 1000 Volts (Insulated, High Temperature)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

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MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals

DRAWINGS

Naval Ordnance Systems Command

2639851	Test Cable Assembly

Bureau of Naval Weapons

LD 290284	Torpedo Control Cable Torpedo Connector Receptacle Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First articles. Unless otherwise specified in the contract or purchase order, three first articles of the test cable assembly shall be manufactured using the methods and procedures proposed for production. The first articles will be tested as specified in section 4 to determine if the proposed production method will yield an item which meets the technical and design requirements of this specification and the contract or purchase order.

3.2 Materials.

3.2.1 Fungus proof. Materials shall be selected in accordance with MIL-STD-454, requirement 4.

3.2.2 Corrosion resistance. Materials shall be corrosion-resistant or suitably treated to resist corrosion.

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3.2.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact.

3.2.4 Nonmagnetic materials. Nonmagnetic materials shall be used where practical.

3.2.5 Fumes. Materials which, under the conditions likely to be encountered in storage or service, liberate deleterious fumes shall not be used.

3.3 Design and construction. The design and construction shall be in accordance with the requirements specified herein and on NAVORD Drawing 2639851.

3.4 Weight. The test cable assembly shall not weigh more than 20 pounds.

3.5 Life. The test cable assembly shall be designed and constructed to withstand a service life of 10 years.

3.6 Strength. The test cable assembly shall withstand without damage axial loads of 200 pounds. The cable portion of the test cable assembly shall withstand without damage 20 60-second applications of 200 pounds perpendicular to the cable axis.

3.7 Mating force. The torpedo plug of the test cable assembly shall mate and unmate with connector receptacle BUWEPS LD 290284 with a force of 30 to 60 pounds.

3.8 Cable construction. The cable shall consist of 65 conductors protected by a 0.080-inch (minimum) thick, molded polychloroprene jacket in accordance with MIL-S-6855, class II, grade 60. The finished cable diameter shall be between 0.925 and 0.965 inch. The jacket shall be free of porosity and release agents. The cable case shall also be free of release agents.

3.8.1 The conductors shall be free of splices, broken wires or strands, and insulation breaks which affect the operation or service life of the test cable assembly.

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3.8.2 The 65 conductors shall be as follows:

(a) Four AWG No. 16 single conductors in accordance with MIL-W-16878/5, type EE-16.

(b) Eight AWG No. 18 single conductors in accordance with MIL-W-16878/5, type EE-18.

(c) Forty-six AWG No. 22 single conductors in accordance with MIL-W-16878/5, type EE-22.

(d) Seven AWG No. 22 shielded single conductors in accordance with MIL-W-16878/5, type EE-22, nominal outside diameter over outer insulating jacket, 0.099 inch.

3.8.3 The jacket of each conductor shall be distinguishable from all other jackets in the cable assembly.

3.8.4 The cable shall have a concentric lay with a pitch of 6 to 8 inches.

3.9 Insulation resistance. With 500 ± 50 volts direct current applied between any two pins or between any pin and either connector shell, insulation resistance shall be 50 megohms minimum.

3.10 Dielectric. The test cable assembly shall withstand 1000 ± 50 volts alternating current, 60 ± 2 hertz applied between any two pins or between any pin and either connector shell for 5 seconds without breakdown.

3.11 Torsion. The test cable assembly shall withstand without failure at least 3000 cycles of torsion, consisting of plus and minus 90° twists made over one 10-inch interval of its length.

3.12 Low temperature bending. The test cable assembly shall withstand without failure at least 5000 bends of 90° of no more than 6-inch radius, over one 12-inch interval of its length, at a temperature of -40° Celsius.

3.13. Hydrostatic pressure. The test cable assembly shall withstand 50 ± 5 pounds per square inch of water pressure without degradation or failure.

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3.14 Contact-conductor circuit resistance. The maximum direct-current resistance of each circuit shall be as follows:

Conductor AWG size	Maximum direct-current resistance
No. 22	0.8 ohm
No. 18	0.5 ohm
No. 16	0.5 ohm

3.15 Environmental requirements. The test cable assembly shall meet the performance requirements of this specification during or after the following environmental conditions:

(a) Vibration, operating: MIL-STD-167, type I equipment
(b) Vibration, nonoperating: 3.5 g at 5 to 500 hertz
(c) Shock, nonoperating: 40 g for 18 milliseconds
(d) High temperature, storage: 85° Celsius
(e) Low temperature, storage: 62° Celsius
(f) Humidity, nonoperating: 95 percent relative at 56° Celsius
(g) Humidity, operating 85 percent relative at 56° Celsius
(h) Salt fog atmosphere, nonoperating.

3.16 Workmanship. The workmanship of the test cable assembly shall be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the test cable assembly shall be classified as follows:

(a) First article inspection
(b) Quality conformance inspection.

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4.2.1 First article inspection. Unless otherwise specified in the contract or purchase order, three test cable assemblies shall be manufactured using the methods and procedures proposed for production. These test cable assemblies shall be subjected to the first article inspection procedure of 4.4.

4.2.2 Quality conformance inspection. The procedure of 4.5 shall be performed on each cable assembly submitted for acceptance under contract (100 percent inspection) to assure that production units meet the requirements of this specification and the contract or purchase order.

4.3 Test conditions. Unless otherwise specified herein, the test cable assembly shall be tested under the following ambient conditions:

(a) Temperature: 25° ± 10° Celsius
(b) Humidity: not exceeding 95 percent relative.

4.4 First article inspection procedure.

4.4.1 First article inspection shall be performed on three first article units in accordance with the test schedule shown in table I.

4.4.2 A failure of any test cable assembly in any test or inspection shall be cause for denial of production approval.

4.5 Quality conformance inspection procedure.

4.5.1 Quality conformance inspection shall be performed on each test cable assembly submitted for acceptance under the contract according to the test schedule shown in table I.

4.5.2 A failure of any test cable assembly in any test or inspection shall be cause for rejection of that assembly.

4.6 Visual and mechanical examination. The test cable assembly shall be examined to verify that the design, construction, physical dimensions, and workmanship are in accordance with the applicable requirements specified herein and NAVORD Drawing 2639851.

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Table I

TEST SCHEDULE

Test	Requirement	Method	First article unit			Quality conformance
			A	B	C	
Visual and mechanical examination	3.2, 3.3, 3.4, 3.7, 3.8, 3.16	4.6	X	X	X	X
Contact-conductor circuit resistance	3.14	4.7.1	X	X	X	X
Insulation resistance	3.9	4.7.2	X	X	X	X
Dielectric strength	3.10	4.7.3	X	X	X	X
Torsion	3.11	4.7.4	X		X	
Low temperature bending	3.12	4.7.5	X		X	
Cable strength	3.6	4.7.6	X	X		
Test cable assembly strength	3.6	4.7.7	X	X		
Hydrostatic pressure	3.13	4.7.8	X		X	
Vibration	3.15	4.7.9	X		X	
Shock	3.15	4.7 10	X	X		
High temperature, storage	3.15	4.7.11	X		X	
Low temperature, storage	3.15	4.7.12	X		X	
Humidity	3.15	4.7.13	X	X		
Salt fog	3.15	4.7.14		X		
Fungus	3.2.1	4.7.15		X		

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4.7 Test methods.

4.7.1 Contact-conductor circuit resistance. The direct current resistance of each of the 65 conductors shall be measured with an ohmmeter having an accuracy of at least ±0.02 ohm.

4.7.2 Insulation resistance. The insulation resistance shall be measured between every pair of conductors and between every conductor and each connector shell, using a device capable of measuring at least 50 megohms with ±5 percent accuracy and which applies 500 ± 50 volts direct current for the measurement.

4.7.3 Dielectric strength. The test cable assembly shall be subjected to a 5-second application of 1000 ± 50 volts alternating current at 60 ± 2 hertz, applied to each conductor sequentially with all adjacent conductors at ground potential, using a device capable of detecting current leakage of 10 microamperes or more with ±5 percent accuracy.

4.7.4 Torsion.

4.7.4.1 The test cable assembly shall be mounted in the test apparatus of figure 1 for the torsion test. The cable shall be clamped at two points on the test fixture such that a 90° ± 5° bend, with a radius not to exceed 4 inches, shall be formed within a 10-inch length of cable (see figure 1). The cable shall be rigidly secured to a fixed support at one clamp position and rigidly secured to a pivot arm at the second clamp position. The pivot axis of the arm shall be an extension of the centerline of the cable which is clamped to the fixed support.

4.7.4.2 One cycle of torsion shall consist of a 90° ± 5° rotation in one direction, a 180° ± 10° rotation in the reverse direction, and a 90° ± 5° rotation in the original direction to the start position, in a plane perpendicular to the fixed support.

4.7.4.3 The test shall consist of 3000 cycles of torsion at the rate of 2 cycles per minute. The time duration of one cycle shall approximate but not exceed 15 seconds. Continuity of all conductors shall be monitored continuously. The test shall be stopped and the number of cycles recorded in the event of failure of any conductor.

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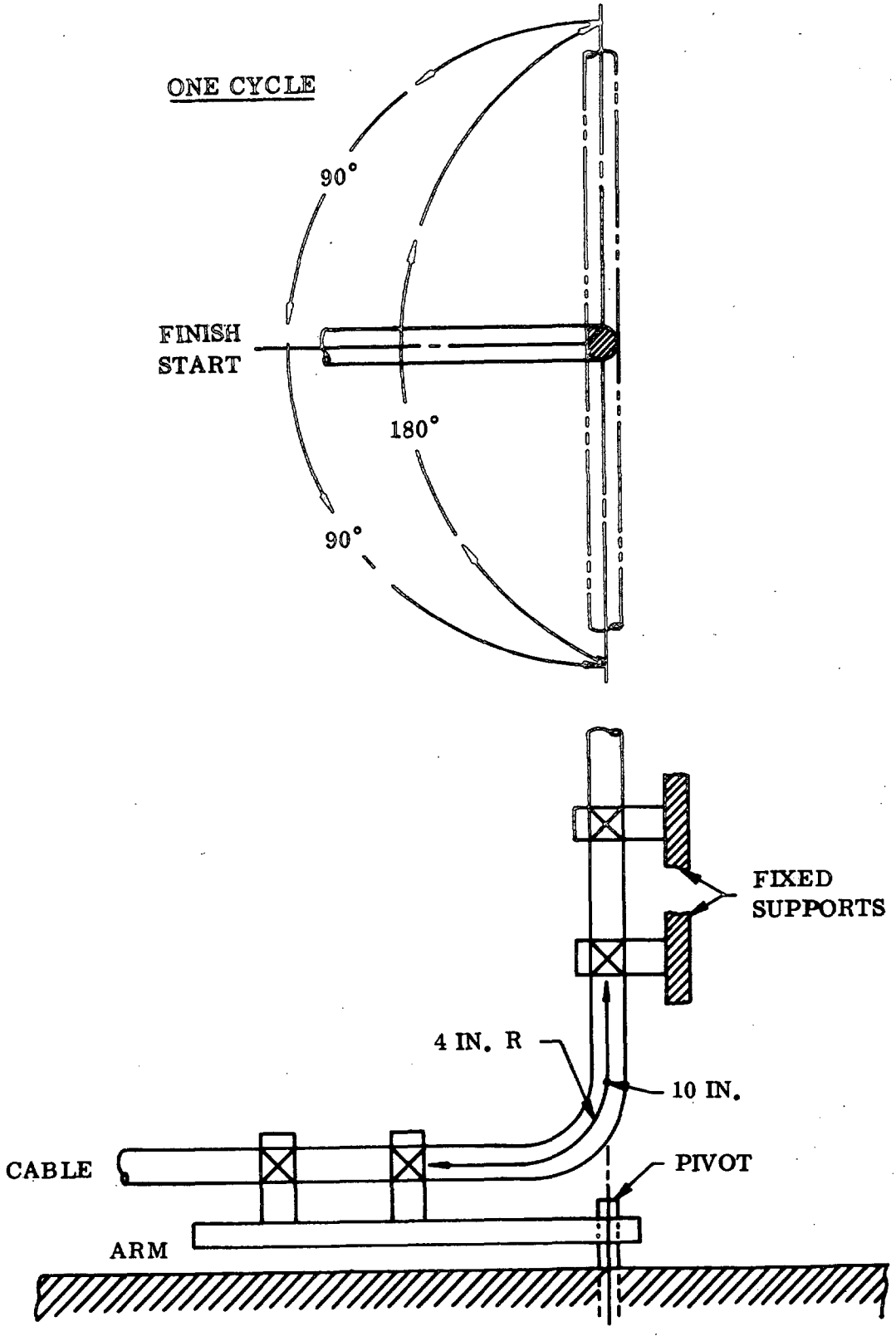


FIGURE 1. TEST SETUP, TORSION TEST

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4.7.5 Low-temperature bending.

4.7.5.1 While the test cable is mounted in the test apparatus of figure 2, a 90° ± 5° bend test shall be performed at -40° ± 3° Celsius. The test cable assembly and test fixture shall be conditioned at the test temperature for at least 12 hours prior to initiation of the test.

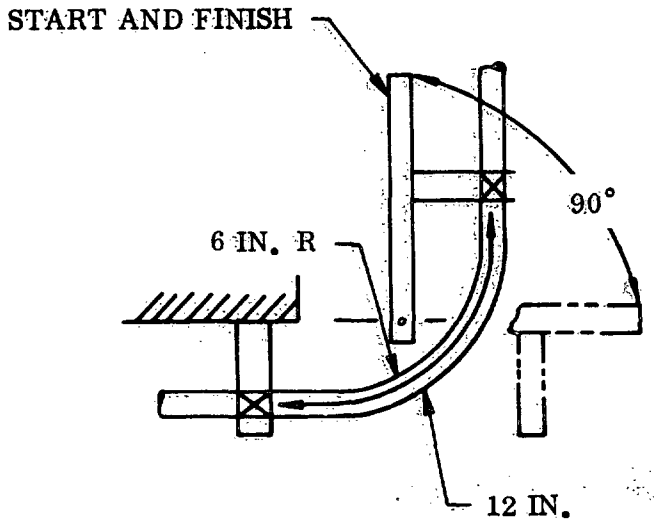


FIGURE 2. TEST SETUP, LOW TEMPERATURE BENDING TEST

4.7.5.2 The cable shall be mounted and clamped at two points on the test fixture such that a 90° ± 5° bend, with a radius not to exceed 6 inches, shall be formed within a 12-inch length of cable (see figure 2). The cable shall be rigidly secured to the underside of a fixed support, lying in a horizontal plane at one clamp position, and rigidly secured to a pivoting arm which shall be mounted to oscillate from a vertical position to a horizontal position. When in its horizontal position, its axis shall form an extension of the plane of the fixed support, and the cable shall be attached to the underside of the arm. The attachment of the cable to the oscillating arm shall be made when the arm is in the vertical position. The pivot axis of the oscillating arm shall intersect the longitudinal axis of the fixed support and shall be equidistant from the two clamping points. The fixture shall be designed so that the section of cable between the clamp positions does not touch any portion of the mechanism or cold box during the test.

4.7.5.3 One cycle shall consist of oscillation of the pivot arm from the vertical position to the horizontal and back to the vertical. The

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test shall be conducted at a rate of 1 cycle per minute with each half cycle to be accomplished within 6 seconds. The delay between cycles may occur at the end of either half cycle or may be divided to occur at the end of each half cycle.

4.7.5.4 The test shall consist of 5000 cycles of bending. Continuity of all conductors shall be monitored continuously. The test shall be stopped and number of cycles recorded in the event of failure of any conductor.

4.7.6 Cable strength. The cable shall be securely clamped at two points along its length 15 inches apart. The equipment shall be oriented such that the axis of the 15-inch length of cable is in a horizontal plane. At a point midway between the clamps, a weight of 200 pounds shall be suspended from the cable for a period of at least 60 seconds, not including the application and removal periods. The shoe or weight support shall be no more than 1-1/2 inches wide and of such design as to spread the 200-pound load over the 1-1/2 inch length of cable. The cable sheath may be protected or padded to spread the load and prevent damage to the cable sheath. The load shall be applied and released a total of 20 times over a period not to exceed 2 hours. At the conclusion of the tests, all conductors in the cable shall be tested for continuity.

4.7.7 Test cable assembly strength. The test cable shall be placed in a test fixture such as that shown in figure 3 and a 200-pound load applied. While under load, the test cable assembly shall be subjected to the procedure of 4.7.1.

4.7.8 Hydrostatic pressure. The simulator plug shall be capped to seal the engaging face. The cable assembly shall be mounted in the hydrostatic pressure tank as shown in figure 4 and the pressure raised to 50 ± 5 pounds per square inch. The pressure shall be maintained for a minimum of 30 minutes. During the last 10 minutes of the test, the insulation resistance test of 4.7.2 shall be performed.

4.7.9 Vibration.

4.7.9.1 Vibration, operating. The test cable assembly, less packaging, shall be subjected to the procedure of 4.7.1 during the test of type I mechanical vibration as specified in MIL-STD-167.

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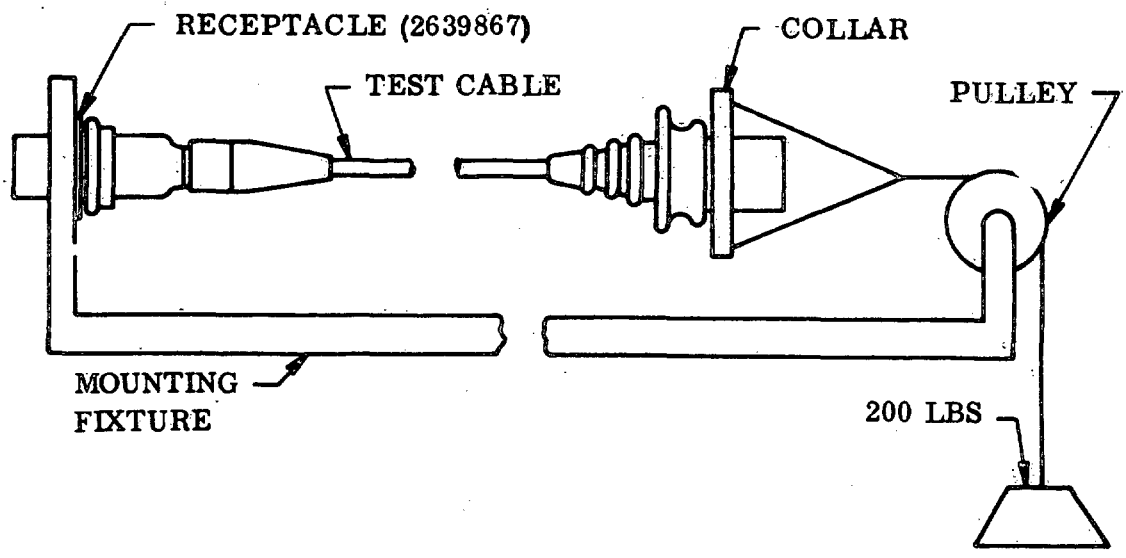


FIGURE 3. TEST CABLE ASSEMBLY, STRENGTH TEST

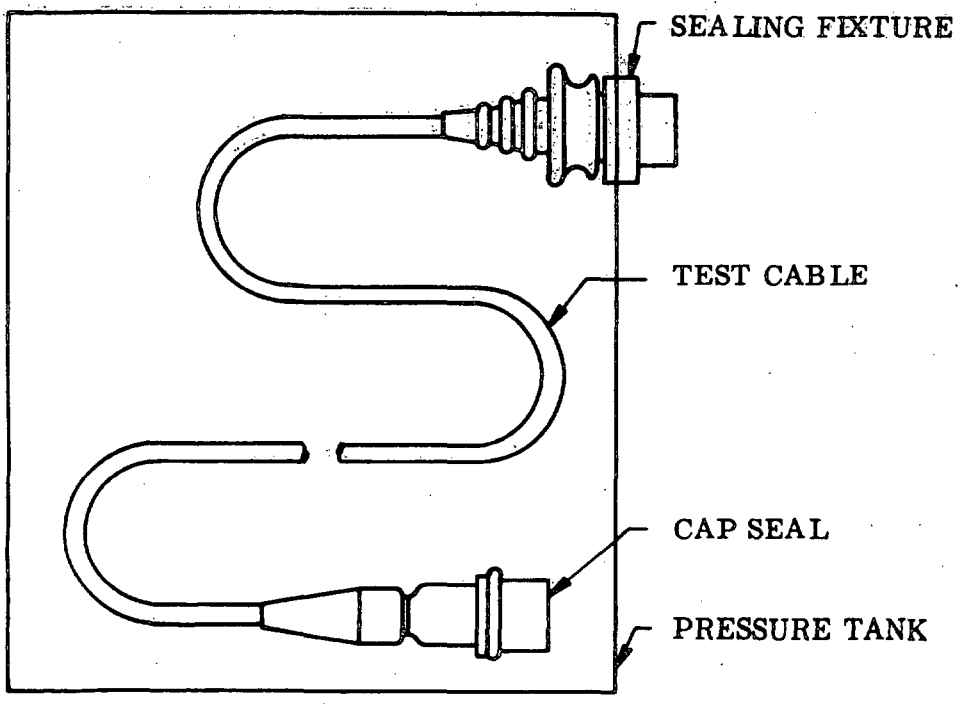


FIGURE 4. HYDROSTATIC PRESSURE TEST

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4.7.9.2 Vibration, nonoperating. The test cable assembly, less packaging, shall be subjected to vibration cycling in accordance with MIL-STD-810, method 514, procedure X, curve AY, for a period of 1 hour in each of the three axes. At the completion of this test the cable shall be examined for physical damage and shall be subjected to the procedures of 4.7.1, 4.7.2, and 4.7.3.

4.7.10 Shock. The test cable assembly, less packaging, shall be subjected to the shock test of MIL-STD-810, method 516, procedure I, figure 516.1-2, amplitude (b), duration (d). At completion of this test the cable shall be examined for physical damage and shall be subjected to the procedures of 4.7.1, 4.7.2, and 4.7.3.

4.7.11 High temperature, storage. The test cable assembly, less packaging, shall be subjected to a temperature of 82.5° ± 2.5° Celsius for a period of 24 hours minimum. After returning to room temperature, the test cable assembly shall be examined for physical damage and shall be subjected to the procedures of 4.7.1, 4.7.2, and 4.7.3.

4.7.12 Low temperature, storage. The test cable assembly, less packaging, shall be subjected to a temperature of -58.5° Celsius for a period of 72 hours minimum. After returning to room temperature, the test cable assembly shall be examined for physical damage and shall be subjected to the procedures of 4.7.1, 4.7.2, and 4.7.3.

4.7.13 Humidity. The test cable assembly, less packaging, shall be tested in accordance with MIL-STD-810, method 507, procedure I, except that the upper temperature shall be 52.7° ± 3.5° Celsius instead of 71° Celsius. Within 1 hour after completion of the test, the cable shall be examined for corrosion or other physical damage and shall be subjected to the procedures of 4.7.1, 4.7.2, and 4.7.3.

4.7.13.1 Humidity operating. If specified by the procuring activity, the cable shall be subjected to the tests of 4.7.1, 4.7.2, and 4.7.3, herein, while at 52.7° ± 3.5° Celsius and 85 percent relative humidity during the last cycle of humidity test of 4.7.13.

4.7.14 Salt fog. The test cable assembly shall be tested in accordance with MIL-STD-810, method 509, procedure I. After test, there shall be no evidence of corrosion, which directly affects operation. After drying and removal of salt deposits, particularly from the connector area, the unit shall be subjected to the procedures of 4.7.1, 4.7.2, and 4.7.3.

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4.7.15 Fungus. If specified by the procuring activity, the test cable assembly shall be tested in accordance with MIL-STD-810, method 508, procedure I. After the test, the test cable assembly shall be examined for evidence of deterioration, corrosion, or fungus growth; the test cable assembly shall be subjected to the procedures of 4.7.1, 4.7.2, and 4.7.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified by the procuring activity, the test cable assembly shall be preserved in accordance with MIL-P-116, method II, and packaged and packed in accordance with MIL-STD-794.

5.2 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Shipment nomenclature for marking shall be as a minimum "TEST CABLE ASSEMBLY, NAVORD Drawing 2639851."

6. NOTES

6.1 Intended use. The test cable assembly is intended for use with Test Set Mark 281 Mod 1 to provide a means of connection to Torpedo Marks 14, 16, 37, 45, and Target Torpedo Mark 27.

6.2 Ordering data. Procurement documents should specify, but not be restricted to, the following:

(a) Title, number, and date of this specification

(b) Test cable assembly drawing number: NAVORD Drawing 2639851

(c) Quantity ordered should include first article inspection samples that will be expended as a result of first article tests

(d) Information concerning inclusion of optional environmental tests and/or exceptions to this specification or other pertinent documents

(e) Level of packaging and packing desired, if other than as specified in 5.1.

Custodian:
Navy - OS

Preparing activity:
Navy - OS
(Project Number 1355-N011)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-82576 (OS), Cable Assembly, Test (for Torpedo Test Set Mark 281 Mod 1)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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